PRE-ALGEBRA (7TH) · UNIT 7: SOLVING EQUATIONS AND INEQUALITIES

Summer Job Savings

Compare summer jobs and savings plans with multi-step equations, inequalities, and interest.

CP1

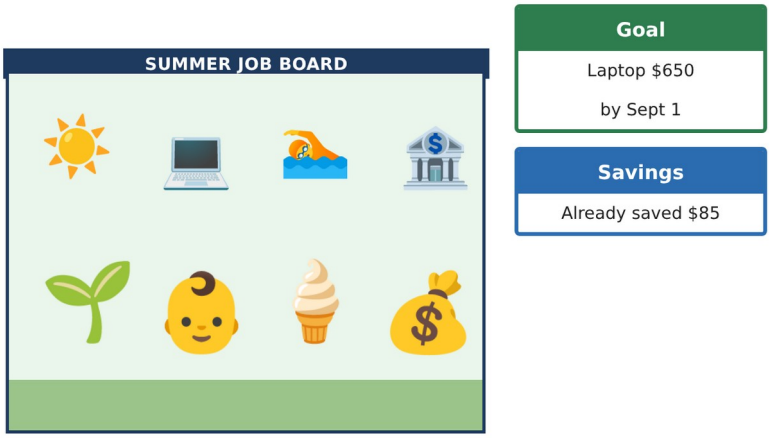
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You want to buy a $650 laptop by the end of the summer. You can mow lawns, babysit, or work at the snack stand at the town pool, and you can put your money in a savings account that earns interest.

You will write and solve equations and inequalities to compare jobs, reach your goal, and plan your savings.



Learning Goals

- I can solve multi-step equations, including variables on both sides.
- I can solve two-step inequalities and interpret the solutions.
- I can transform formulas and compute simple and compound interest.

Project Timeline

When	What you will do
Day 1	Compare Jobs
Day 2	Savings Goal
Day 3	Interest
Final	Summer Savings Plan + reflection

Part 1: Compare Jobs

Day 1 · Use the Job Offer Cards.

Job Offer Cards

Lawn Mowing	Babysitting	Snack Stand
$15 per lawn	$12 per hour	$10.50 per hour
$40 for a mower tune-up	$20 CPR class first	+ $25 signing bonus

Dog Walking	Bank A	Bank B
$8 per walk	3% simple interest	2.8% compounded
no start-up cost	per year	yearly

1. After how many hours do babysitting and the snack stand pay the same total? Solve $12h - 20 = 10.50h + 25$.

2. Which job pays more for 20 hours? For 40 hours?

3. A mowing weekend: you mow l lawns on Saturday and 3 more on Sunday, earning $15(l + l + 3) - 40 = 365$. Solve for l.

4. Solve $\frac{2}{3}(h - 6) = 18$ and explain a babysitting situation it could model.

Part 2: Savings Goal

Day 2 · Goal $650. You have $85.

5. Write and solve an inequality for the number of babysitting hours needed.

6. You can work at most 10 hours a week at the snack stand. Write an inequality for the number of weeks to reach $650 and solve it.

7. You will spend $15 per week on snacks while working. Rewrite and solve the inequality from the last question.

8. Explain how you knew whether to round up or down in each answer.

Part 3: Interest

Day 3 · Simple interest: $I = Prt$. Compound interest: $A = P(1 + r)^t$.

9. Solve $I = Prt$ for r. What rate gives \$24 interest on \$400 in 2 years?

10. Compare \$650 for 3 years in Bank A (3% simple) and Bank B (2.8% compounded yearly).

11. Write your summer plan: job(s), hours or lawns, weeks, and where to save the extra money. Support each choice with an equation or inequality.

Final Product: Summer Savings Plan

Your final product must include:

- Job comparison using variables on both sides
- Savings inequality with rounding decisions
- Transformed interest formula
- Bank comparison
- Complete summer plan

Reflection

R1. What does the solution h = 30 mean when comparing two jobs?

R2. Why might compound interest beat simple interest over many years even with a lower rate?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Equations and inequalities model each job and goal correctly and support a realistic savings plan. Goes beyond the minimum with a creative, realistic choice.	Equations and inequalities model each job and goal correctly and support a realistic savings plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes the job choice, solved equations and inequalities, interest comparison, and a calendar. Polished, organized, and visually engaging.	Plan includes the job choice, solved equations and inequalities, interest comparison, and a calendar. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.